

UNIT-I

Cloud Computing at a Glance: The vision of cloud computing: Defining a cloud A closer look, The cloud computing reference model Characteristics and benefits Challenges ahead

Historical Developments: Distributed systems, Virtualization, Web2.0 Service- oriented computing Utility-oriented computing

Building Cloud Computing Environments: Application development Infrastructure and system development, Computing platforms and technologies.

Important Questions:

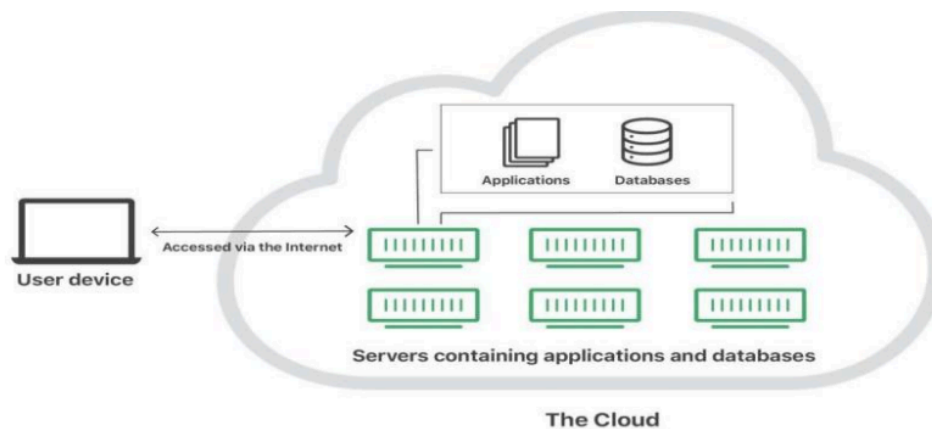
Short Questions(2 Marks each)

- 1) What are the key characteristics of cloud computing?
- 2) Mention two major benefits of cloud computing.
- 3) Explain the features of cloud computing?
- 4) Define Web 2.0 and its role in cloud computing.
- 5) List out the different types of clouds?
- 6) Write any two examples of SaaS.

Long Questions (8 Marks Each)

- 1) Explain the Cloud Computing Reference Model with a diagram and discuss its layers.
- 2) Explain characteristics and benefits of cloud technologies.
- 3) Defining overview architecture of cloud technologies.

The **cloud** refers to internet-based servers and the software or databases running on them. These cloud servers are housed in global data centers, allowing users and businesses to access computing resources without maintaining physical infrastructure.



How the Cloud Works

Instead of relying on local devices for storage and processing, cloud computing enables users to access applications and data from anywhere. For example:

- Logging into an **Instagram** account on a new phone retains all photos, videos, and messages.
- Cloud-based services like **Gmail, Microsoft 365, Dropbox, and Google Drive** store and sync files across multiple devices.

Benefits for Businesses

- **Cost Savings** – Eliminates the need for maintaining physical servers.
- **Scalability** – Companies can easily scale resources based on demand.
- **Global Accessibility** – Employees and customers can access the same services from any location.

Cloud Computing at a Glance

1. **Definition:** Delivers computing services (storage, processing, networking) over the internet.
2. **Resource Access:** Users remotely utilize computing power, storage, and applications.
3. **On-Demand Services:** Provides flexible, scalable, and cost-effective solutions.
4. **Internet-Based:** Eliminates the need for local infrastructure, making services accessible globally.
5. **Payment Model:** Operates on a pay-as-you-go basis, optimizing cost efficiency.
6. **Service Models:**
 - **IaaS (Infrastructure as a Service)** – Virtualized computing resources.
 - **PaaS (Platform as a Service)** – Development and deployment platforms.
 - **SaaS (Software as a Service)** – Cloud-hosted applications for end users.
7. **Shared Environment:** Uses virtualization for enhanced accessibility, collaboration, and innovation.

Definition of Cloud Computing

Cloud computing refers to the **delivery of computing services**—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet. It provides a **virtualized platform** with on-demand resources, dynamically provisioning hardware, software, and data as needed.

Vision of Cloud Computing

1. **Global Accessibility:** Enables users to connect and access services from anywhere.
2. **Scalability:** Allows seamless resource scaling to meet fluctuating demands.
3. **Cost Efficiency:** Operates on a **pay-as-you-go** model, reducing upfront costs.
4. **Innovation Acceleration:** Provides a flexible platform for rapid development and deployment of applications.
5. **Collaboration:** Enhances teamwork through shared resources and global connectivity.

Cloud Computing vs. On-Premises Data Centers

Cloud computing offers an **alternative** to traditional on-premises data centers, reducing the burden of hardware management and infrastructure maintenance.

On-Premises Data Centers

Managing an on-premises data center requires:

- ✓ Purchasing and installing hardware
- ✓ Setting up virtualization and operating systems
- ✓ Installing applications and configuring networks
- ✓ Managing security (firewalls) and storage
- ✓ Continuous maintenance throughout its lifecycle

Cloud Computing Advantage

With cloud computing, the **cloud vendor** handles:

- ✓ Hardware purchase and maintenance
- ✓ Virtualization, storage, and security
- ✓ Software and platform services on demand

Users can **rent** only the required services, paying based on actual usage. The cloud provides an easily accessible **online portal** for managing compute, storage, network, and application resources efficiently.

NIST's Five Essential Characteristics of Cloud Computing

The **National Institute of Standards and Technology (NIST)** defines five core characteristics of cloud computing:

1. **On-Demand Self-Service:** Users can provision resources without requiring human intervention from the provider.
2. **Broad Network Access:** Services are accessible from **any device** over the internet.
3. **Resource Pooling:** Computing resources are shared among multiple users through **multi-tenancy**.
4. **Rapid Elasticity:** Resources can be automatically scaled up or down as needed.
5. **Measured Service:** Usage is **monitored and billed** based on actual consumption.

Basic Concepts of Cloud Computing

Cloud computing is made possible by **various services and models** that work behind the scenes to provide a seamless experience for end users. These include:

- **Deployment Models** – Defines how the cloud is accessed.

- **Service Models** – Specifies the type of cloud services provided.

Deployment Models

Deployment models determine how cloud resources are accessed and managed. There are **four** primary types:

1. Public Cloud

- **Off-site & remote:** Resources are hosted and managed by a **third-party provider**.
- Users access services **on-demand** over the **internet** via APIs or web applications.
- **Cost-efficient** – Billed on a **pay-as-you-go** model (utility computing).
- Examples: **AWS, Microsoft Azure, Google Cloud**



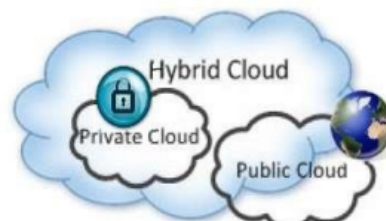
2. Private Cloud

- **Dedicated** cloud infrastructure for a **single organization**.
- Often used as a **first step** before adopting public cloud solutions.
- Provides **enhanced security, control, and compliance**.
- Hosted **on-premises** or by a **third-party** exclusively for the organization.



3. Hybrid Cloud

- **Combination** of **on-premises infrastructure** and **public cloud** services.
- Allows businesses to **keep sensitive data on-premises** while leveraging **cloud solutions** for other functions.
- Common uses: **Disaster recovery, backups, development & testing environments**.



4. Community Cloud

- Shared infrastructure used by multiple **organizations with similar requirements**.
- Offers **greater security** and cost-sharing benefits compared to a public cloud.
- Used in sectors like **government, healthcare, or financial institutions** where data privacy is crucial.



Explain the Cloud Computing Reference Model with a diagram and discuss its layers

Cloud Service Models

Cloud computing is categorized into three primary service models:

1. Infrastructure-as-a-Service (IaaS)

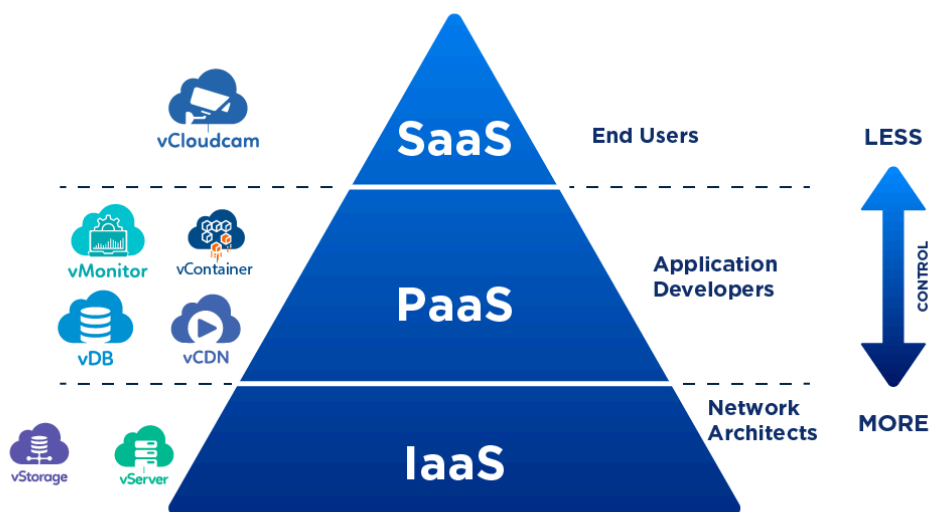
- **Definition:** The service provider owns and manages **hardware resources** such as servers, storage, and networks, which are offered to clients on a pay-per-use basis.
- **Key Features:**
 - Provides **virtualized computing resources** over the internet.
 - Users can **rent** computing infrastructure instead of investing in hardware.
 - Highly **scalable** and **cost-efficient**.
- **Examples:** Amazon EC2, Amazon S3, Microsoft Azure Virtual Machines

2. Platform-as-a-Service (PaaS)

- **Definition:** Offers a **complete development environment** that includes the necessary tools for designing, developing, testing, deploying, and hosting applications—without managing underlying infrastructure.
- **Key Features:**
 - Provides **pre-configured development platforms**.
 - Developers can focus on coding without worrying about **hardware and system maintenance**.
 - Supports **collaborative development** and **faster application deployment**.
- **Examples:** Google App Engine, Microsoft Azure App Services

3. Software-as-a-Service (SaaS)

- **Definition:** The service provider delivers **software applications over the internet**, eliminating the need for installation, maintenance, upgrades, and licensing on local machines.
- **Key Features:**
 - Accessible from **any device with an internet connection**.
 - No need for **manual software installation or updates**.
 - Provides **scalability** and **multi-user collaboration**.
- **Examples:** Google Docs, Salesforce CRM, Microsoft Office 365
- **Google Docs** – A cloud-based document editing tool that allows users to **collaborate in real-time**.
- **Dropbox** – A file storage and sharing service that enables users to **store and access files from anywhere**.



Characteristics of Cloud Technologies

1. **Agility** – Organizations can quickly adapt to changing demands with scalable cloud resources.
2. **Cost Efficiency** – Centralized infrastructure reduces costs by leveraging locations with lower operational expenses.
3. **Device & Location Independence** – No hardware maintenance is required; users can access services from any device, anywhere.
4. **Pay-Per-Use Model** – Users only pay for the resources they consume, improving utilization and reducing waste.
5. **Performance Monitoring** – IT experts from the service provider continuously monitor and optimize performance.
6. **Increased Productivity** – Multiple users can collaborate on the same data simultaneously, improving efficiency.
7. **Time Savings** – Automatic data matching eliminates redundant data entry.
8. **High Availability & Reliability** – Redundant sites ensure minimal downtime and continuous service availability.

9. **Scalability & Elasticity** – Resources are provisioned dynamically in near real-time to meet demand without over-engineering.
10. **Self-Service Interface** – Users can manage cloud resources independently through intuitive management tools.
11. **Virtualization** – Resources are abstracted and virtualized, enhancing flexibility and efficiency.
12. **Enhanced Security** – Centralized data storage improves security with encryption, monitoring, and compliance measures.

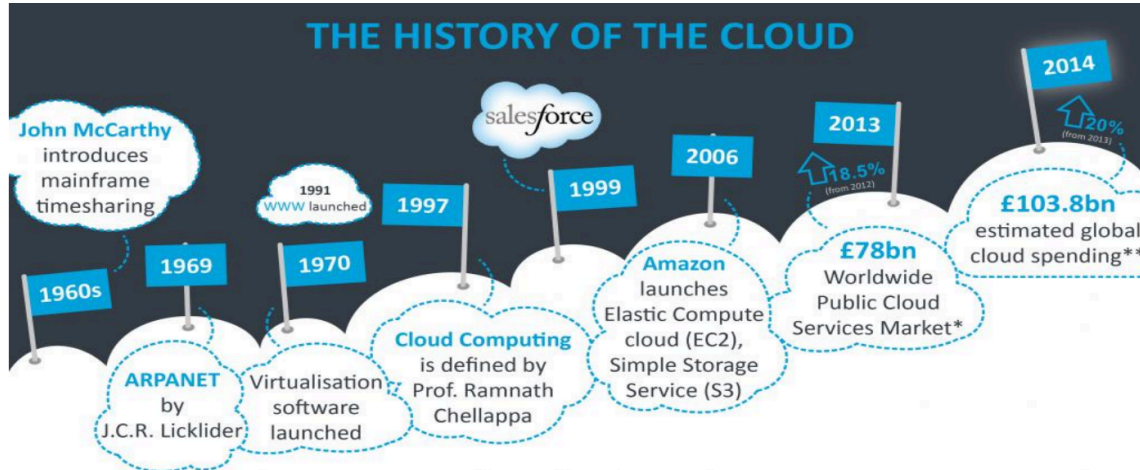
Benefits of Cloud Technologies

1. **Cost Efficiency** – Eliminates the need for purchasing expensive hardware and software, reducing capital expenditure.
2. **Speed & Accessibility** – Resources are available within minutes, making cloud computing highly responsive.
3. **Scalability** – Easily scale resources up or down based on business needs.
4. **Increased Productivity** – Reduces operational efforts since maintenance, patching, and updates are handled by the cloud provider.
5. **Reliability** – Fast and cost-effective data backup and recovery ensure business continuity.
6. **Enhanced Security** – Cloud providers implement advanced security measures, including encryption, monitoring, and compliance policies.

Challenges Ahead in Cloud Computing

1. **Security & Privacy Concerns** – Data breaches and unauthorized access remain risks.
2. **Compliance & Legal Issues** – Meeting regulatory requirements across different regions.
3. **Downtime & Service Reliability** – Cloud outages can impact business operations.
4. **Data Portability & Vendor Lock-in** – Moving data between cloud providers can be difficult.
5. **Network Dependency** – High reliance on internet connectivity for access.
6. **Performance Variability** – Shared infrastructure may impact performance during peak times.
7. **Hidden Costs** – Unexpected expenses from data transfers, scaling, and premium services.
8. **Lack of Control** – Businesses have less direct control over infrastructure and updates.
9. **Skill Gaps** – Organizations need skilled professionals for cloud management.
10. **Integration Challenges** – Connecting cloud systems with legacy applications can be complex.

History & Evolution of Cloud Computing

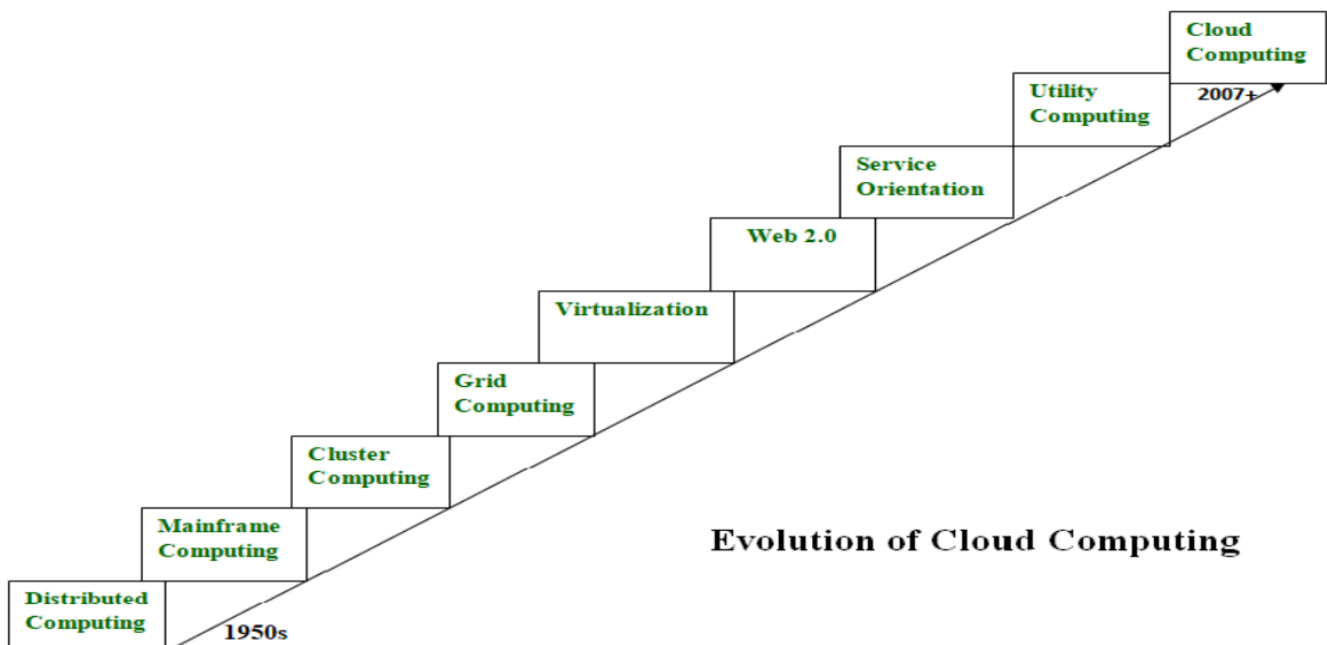


Evolution of Cloud Computing

Cloud computing has evolved through various **technological advancements**, starting from **distributed systems** to modern-day **AI-driven cloud services**.

Key Stages in Cloud Evolution

1. **Distributed Systems**
2. **Virtualization**
3. **Web 2.0**
4. **Service-Oriented Computing**
5. **Utility-Oriented Computing**



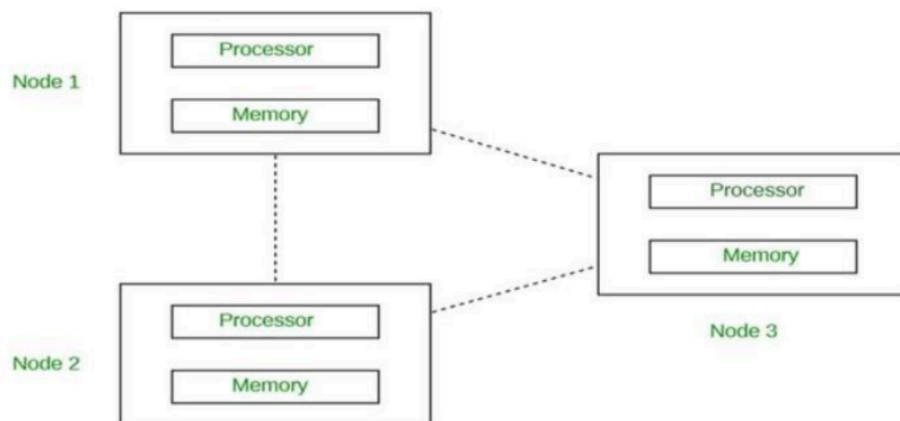
Evolution of Cloud Computing

1. Distributed Systems

A **distributed system** is a network of independent computers that function as a single system. It provides **scalability, reliability, and fault tolerance**.

Distributed Computing

- Forms the fundamental model for cloud computing as cloud environments are inherently distributed.
- Conceals the complexity of cloud systems, providing a unified interface to users.
- Involves multiple computer systems collaborating on a single problem by breaking it into smaller sub-problems.
- Enhances performance, efficiency, and fault tolerance.
- Communication occurs over a network where each processor has local memory.



Properties of Distributed Systems:

- ✓ **Heterogeneity** – Different types of computers & networks working together.
- ✓ **Scalability** – Can grow as needed.
- ✓ **Transparency** – Users don't see underlying complexities.
- ✓ **Independent Failure** – One node failure doesn't affect the whole system.

Milestones in Distributed Computing:

1. **Mainframe Computing**
2. **Cluster Computing**
3. **Grid Computing**

2. Mainframe Computing (1950s-1980s)

- Introduced in **1951**, mainframes were **powerful & reliable computing machines**.

- Used for **bulk data processing, online transactions, and enterprise resource planning (ERP)**.
- **Expensive & centralized**, requiring dedicated operators.
- Applications: **Banking, airline ticketing, government services**.

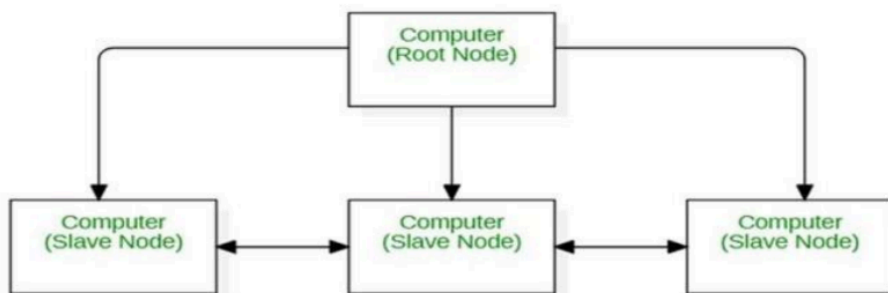
3. Cluster Computing (1980s-1990s)

- **Multiple computers connected together** to act as a single system.
- More **cost-effective & scalable** than mainframes.
- **Easier to expand** – new nodes could be added as needed.
- Used in **databases, web servers, and enterprise applications**.

A **cluster** is a collection of independent computers, known as **nodes**, that work together to execute tasks as a single system. In **cluster computing**, two or more nodes collaborate to improve **performance, scalability, and efficiency**.

The primary goal of cluster computing is to enhance system capabilities by distributing workloads across multiple machines. Each node, regardless of whether it is a parent or child node, functions as part of a unified entity to process tasks efficiently.

This setup allows clusters to handle large-scale computations, database operations, and high-performance applications while ensuring **fault tolerance and scalability**.



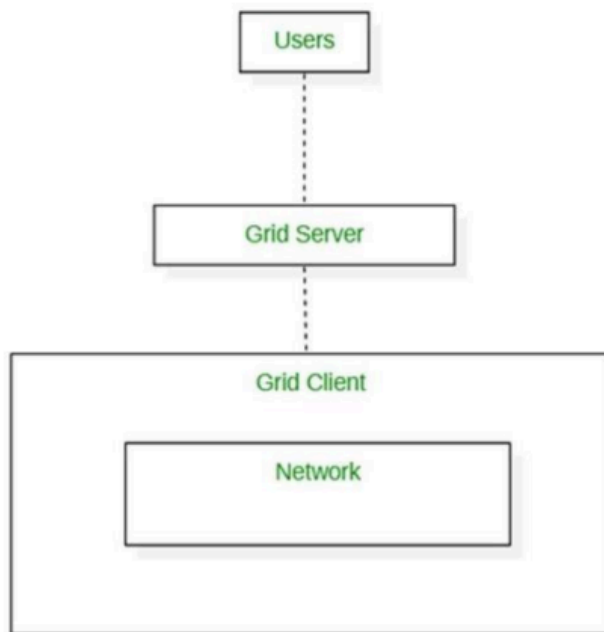
4. Grid Computing (1990s-2000s)

- A **global network of computers** working together as a **virtual supercomputer**.
- Used for **high-performance computing, simulations, AI research**.
- Grid computing is often considered the **precursor to cloud computing**.
- Applications: **Scientific research, predictive modeling, automation**.
- **Challenges**: Complex networking & high maintenance costs.

Grid computing is a type of computing where multiple computers are connected to work together as a **virtual supercomputer**. This system helps process tasks that are too complex for a single machine.

Key Points:

- Used for tasks requiring **high computing power** and handling **large datasets**.
- Computers in the grid communicate through a **data grid** to share resources efficiently.
- **Goal:** Solve complex problems faster and improve productivity by distributing workloads across multiple machines.



5. Virtualization (2000s-Present)

- The **core technology behind cloud computing**.
- Creates **virtual machines (VMs)**, allowing multiple OSs to run on the same hardware.
- Enabled by tools like **VMware, Xen, Hyper-V**.
- **Key Benefit: Optimized resource utilization**, leading to **Infrastructure-as-a-Service (IaaS)** models like AWS EC2.
- Virtualization abstracts physical hardware or runtime environments.
- It creates a virtual layer over hardware, allowing multiple instances to run simultaneously.
- It serves as the foundation for cloud computing services like Amazon EC2 and VMware vCloud.

6. Web 2.0 (2004-Present)

- Provides the interface to access cloud services.
- Enables interactive and dynamic web applications.
- Supports the Everything-as-a-Service (XaaS) model.

Web 2.0 refers to the second generation of the internet, characterized by **interactive, user-generated content, and social networking**. Unlike **Web 1.0**, which was static, Web 2.0 allows users to collaborate, share, and contribute content dynamically.

Role of Web 2.0 in Cloud Computing

- **Enables On-Demand Services:** Web 2.0 technologies, such as AJAX and RESTful APIs, allow seamless integration with cloud services.
- **Supports SaaS Applications:** Many cloud-based SaaS applications (e.g., Google Docs, Dropbox) rely on Web 2.0 principles for accessibility and collaboration.
- **Enhances User Experience:** Web 2.0 applications leverage cloud computing to provide real-time updates, data synchronization, and rich internet applications (RIA).
- **Facilitates Social and Collaborative Platforms:** Platforms like Facebook, YouTube, and Twitter use cloud computing to store and process vast amounts of user-generated content.
- Marked the shift from **static web pages** to **interactive web applications**.
- Enabled **dynamic, real-time content** and **user-generated data**.
- Examples: **Google Maps, Facebook, Twitter, YouTube**.
- **Cloud computing became essential** for storing & processing massive amounts of user data.

7. Service-Oriented Computing

- Introduced **Service-Oriented Architecture (SOA)**, where applications were designed as **independent services**.

Service Orientation

- Acts as a reference model for cloud computing.
- Supports cost-effective, flexible, and scalable applications.
- **Infrastructure as a Service (IaaS):** Dynamically scales resources up or down.
- **Platform as a Service (PaaS):** Embeds code with algorithms and rules.
- **Quality of Service (QoS)** and **Service Level Agreements (SLA)** were introduced.
- Led to the development of **Software-as-a-Service (SaaS)** models.
- Examples: **Salesforce, Google Workspace, Microsoft Office 365**.

8. Utility Computing (Pay-per-use Model)

- Cloud computing adopted a **pay-per-use model**, similar to public utilities (electricity, water).
- Resources like **storage, compute power, and bandwidth** became **on-demand**.

- Key Cloud Providers: **AWS, Google Cloud, Microsoft Azure.**

Advantages of Utility Computing

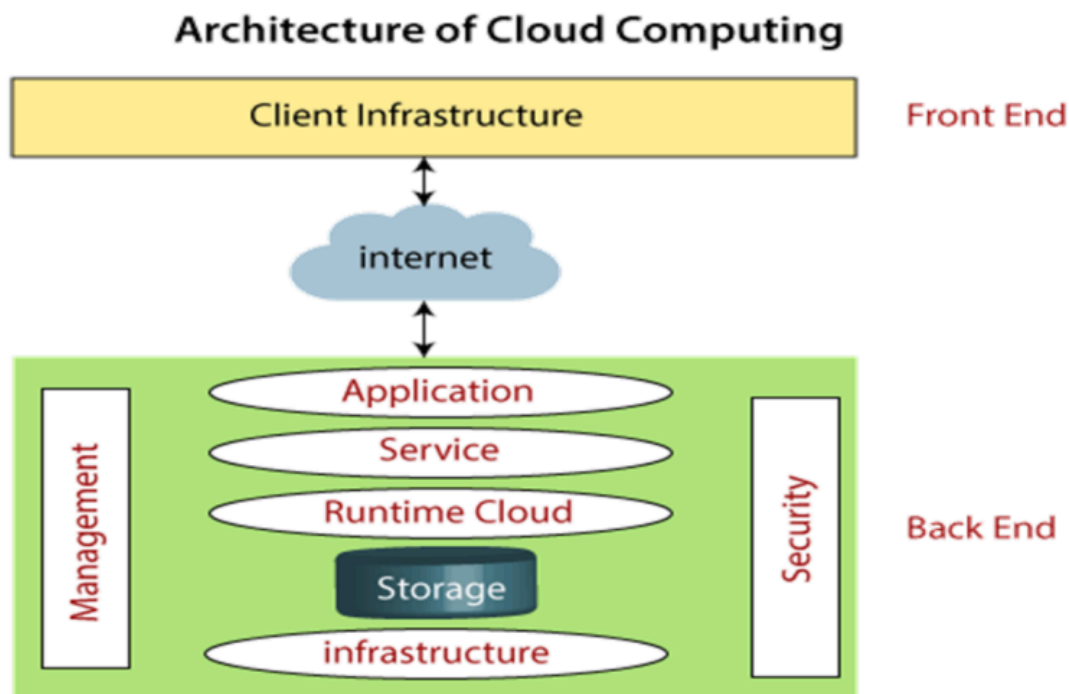
- ✓ **Cost-Efficient** – Pay only for what you use.
- ✓ **Scalability** – Easily scale resources up or down.
- ✓ **Flexibility** – Access resources from anywhere.

Disadvantages

- ✗ **Internet Dependency** – Requires a strong network connection.
- ✗ **Vendor Lock-in** – Migration between cloud providers is difficult.
- ✗ **Security Concerns** – Data stored remotely can be vulnerable.

Architecture of Cloud Computing

Cloud computing architecture integrates **Service-Oriented Architecture (SOA)** and **Event-Driven Architecture (EDA)** to provide scalable, flexible, and on-demand services. It consists of various components such as **client infrastructure, application, services, runtime cloud, storage, infrastructure, management, and security.**



The architecture of cloud computing is divided into two main parts:

1. Frontend (Client-Side)

- Used by **end-users** to interact with cloud services.
- Includes client-side interfaces such as **web browsers (Chrome, Firefox, Edge), mobile apps, and tablets**.
- Provides an interface for accessing cloud-based applications and services.

2. Backend (Provider-Side)

- Managed by **cloud service providers**.
- Controls resources required to provide cloud services, including **storage, virtual machines, databases, security mechanisms, and servers**.
- Includes various computing models such as **IaaS, PaaS, and SaaS**.

Key Components of Cloud Computing Architecture

1. Client Infrastructure

- It is a **frontend component** that allows users to interact with the cloud using a **Graphical User Interface (GUI)**.
- Example: **Web browsers, mobile apps, virtual desktops**.

2. Application

- Software or platforms that users can access in the cloud.
- Example:
 - **Microsoft 365 (SaaS)**: Provides Word, Excel, and PowerPoint via the cloud.
 - **Gmail**: A cloud-based email service.

3. Cloud Services (IaaS, PaaS, SaaS)

- **Infrastructure as a Service (IaaS)**: Provides virtualized computing resources over the internet.
 - Example: **AWS EC2, Microsoft Azure Virtual Machines**
- **Platform as a Service (PaaS)**: Offers a development platform for building applications.
 - Example: **Google App Engine, Heroku**
- **Software as a Service (SaaS)**: Delivers fully functional applications to end-users.
 - Example: **Dropbox, Gmail, Google Docs**

4. Runtime Cloud

- Provides the **execution environment** for virtual machines and applications.
- Example: **AWS Lambda (serverless execution of functions)**.

5. Storage

- Offers **high-capacity storage** for data and applications.

- Example: **Amazon S3 (object storage), Google Cloud Storage.**

6. Cloud Infrastructure

- Includes **servers, networking, virtualization, and storage** that power the cloud.
- Ensures availability and scalability.

7. Management

- Responsible for managing **resources, security, and performance monitoring.**
- Ensures **efficient allocation of computing resources.**

8. Security

- Implements encryption, identity management, and compliance measures to **protect user data.**
- Example: **Azure Security Center, AWS Identity & Access Management (IAM).**

9. Internet

- Acts as a **communication bridge** between the frontend and backend.

Building Cloud Computing Environments

Cloud computing environments are created by developing applications, frameworks, platforms, and infrastructure to deliver cloud services.

Application Development in Cloud Computing

Cloud applications take advantage of on-demand scalability and flexibility.

- **Web Applications** – Benefit the most from cloud computing due to their scalability and availability.
- **Web 2.0 for Development** – The internet has become a platform for modern applications.
- **Types of Applications:**
 - **Enterprise Applications** (e.g., ERP software)
 - **Resource-Intensive Applications**
 - **Data-Intensive** – Requires large-scale data processing.
 - **Compute-Intensive** – Needs significant computational power (e.g., scientific research).

- **Renting Cloud Infrastructure** – Instead of purchasing expensive hardware, businesses can rent cloud resources as needed.

Cloud Computing Benefits for Application Development

1. **On-Demand Scalability** – Resources scale automatically as demand increases.
2. **Infrastructure as a Service (IaaS)** – Rent computing power, storage, and networking.
3. **Platform as a Service (PaaS)** – Provides a development environment for applications.
4. **Software as a Service (SaaS)** – Fully hosted applications that behave like desktop applications but run in the cloud.

Cloud Infrastructure Development

- **Cloud computing enables organizations to rent compute power, storage, and network resources** instead of maintaining physical hardware.
- Provides **on-demand and dynamic scaling** across computing layers.
- Key capabilities:
 - Renting compute power (AWS EC2, Google Compute Engine)
 - Scalable storage (Amazon S3, Google Cloud Storage)
 - Virtualized networking (Azure Virtual Network)

Key Components of Infrastructure Development

1. **Distributed Computing**
2. **Virtualization**
3. **Service Orientation**
4. **Web 2.0**

System Development in Cloud Computing:

- Involves creating cloud-based frameworks and platforms.
- Focuses on security, scalability, and cost-efficiency.
- Uses **Infrastructure as Code (IaC)** for automation (e.g., Terraform, AWS CloudFormation).
- Supports hybrid and multi-cloud deployments for flexibility.

Computing Platforms and Technologies in Cloud Computing

Cloud computing platforms provide computing, storage, and networking services for various applications. Here are some major platforms and technologies:

1. Amazon Web Services (AWS)

- **Launched:** 2006

- **Type:** Infrastructure-as-a-Service (IaaS)
- **Key Services:**
 - **EC2:** Virtual machines with customizable hardware.
 - **S3:** Scalable storage for data.
 - **RDS, DynamoDB:** Managed databases.
- **Benefits:**
 - Scalable and secure global infrastructure.
 - Pay-as-you-go pricing.
 - Supports web hosting, AI, IoT, and big data applications.

2. Google App Engine

- **Launched:** 2011
- **Type:** Platform-as-a-Service (PaaS)
- **Key Features:**
 - Automatic scaling of resources.
 - Supports Python, Java, and Node.js.
 - Integrates with Google Cloud services (BigQuery, Firebase).
- **Use Case:** Ideal for web application development.

3. Microsoft Azure

- **Type:** Offers IaaS, PaaS, and SaaS solutions.
- **Key Features:**
 - Hybrid cloud support (Azure Arc).
 - Strong integration with Microsoft tools (Office 365, Active Directory).
 - Includes networking, caching, and content delivery services.

4. Hadoop

- **Type:** Open-source big data framework.
- **Core Components:**
 - **HDFS:** Distributed storage.
 - **MapReduce:** Processing framework.
- **Key Features:**
 - Scalable and fault-tolerant.
 - Includes Hive (SQL queries), Pig (data analysis), and Spark (fast processing).
- **Use Case:** Large-scale data processing.

5. Force.com & Salesforce.com

- **Force.com (PaaS):** Used for custom app development within Salesforce.
- **Salesforce.com (SaaS):** A cloud-based CRM solution.
- **Key Features:**
 - Cloud-based customer management.

- Customizable with Apex and Lightning components.
- Integrates with business workflows.

6. Manjrasoft Aneka

- **Type:** Cloud platform for distributed application development.
- **Key Features:**
 - Supports multiple computing models (Task, Thread, MapReduce).
 - Works with public, private, and hybrid clouds.
 - User-friendly UI and APIs.

Other Cloud Platforms

Platform	Features
Heroku	PaaS for Ruby on Rails apps, auto-scales, high availability.
Google App Engine	Runs Python & Java apps, auto-scaling.
Microsoft Azure	Supports .NET, Java, Ruby; offers authentication & load balancing.
Force.com	PaaS for building enterprise apps with Apex & Visualforce.